

Corrigendum: The Science of Visual Data Communication: What Works

Psychological Science in the
Public Interest

1–2

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Two of the figures in this article and several of the references to the figures have been corrected. These changes are detailed below, and the new figures appear at the end of the text.

On p. 116, in the last sentence of the fourth full paragraph, the callout for Figure 2 has been changed to “(such as that shown in Fig. 2, top of second column).”

On p. 120, Figure 5 has been replaced because the color-blindness model that was used to test the originally published version of the figure was incorrect. The following sentence has been added to the end of the Figure 5 caption for clarification: “Protanopia was simulated in Photoshop.” In addition, in the last paragraph of the left column, the second and third sentences have been updated as follows:

For example, the first row in Figure 5 shows a scatterplot encoded with two colors, green and orange. People with typical vision can see that the green dots have a steep positive correlation and the orange dots make a flat line.

On p. 140, in the first paragraph in the right column, two sentences (the sentence beginning “For example, when viewing. . .” and the sentence following it) have been replaced to more accurately describe Figure 20. The new sentences read as follows:

For example, when viewing results of an evaluation of a new intervention (e.g., a special type of equipment one might use to improve one’s score in a sports competition) relative to a control, one might wonder how much investing in the new equipment is likely to help increase one’s score for a randomly drawn instance of the game. When shown frequentist 95% confidence intervals representing uncertainty on bars representing the control (“Standard Boulder”) and treatment

(“Special Boulder”) outcomes in a fictional boulder-sliding competition (Hofman et al., 2020; see Fig. 20, left), lay viewers were inclined to pay more for the treatment and overestimate effect size relative to when they saw error bars showing a predictive interval representing uncertainty in the underlying measurements (Fig. 20, middle).

On p. 141, the second and third sentences in the caption for Figure 20 have been changed to more accurately describe the figure. The new sentences read as follows:

Hofman et al. (2020) found that participants in an online study who viewed the chart on the left, showing a range extending 1.96 *SE* above and below the mean, were more likely to pay more for a new intervention (i.e., “Special Boulder”) compared with the standard intervention (i.e., “Standard Boulder”) and to overestimate the size of the new intervention’s effect than those who saw the chart in the center, showing a range extending 1.96 *SD* above and below the mean. The chart on the right is the same as the chart on the left, except that it has the *y*-axis range of the chart in the center. This display slightly reduced overpayment and overestimation, which suggests that viewers are sensitive to superficial visual cues, such as how big a difference looks within the frame of the chart.

On p. 143, Figure 23 has been replaced to correct the color palette.

Finally, on p. 148, the figure callout at the end of the fourth full paragraph has been updated to “(see Fig. 24; see also Garcia-Retamero & Galesic, 2009),” and the figure callout at the beginning of the sixth full paragraph has been updated to “(as in Fig. 24; Garcia-Retamero & Cokely, 2017).”

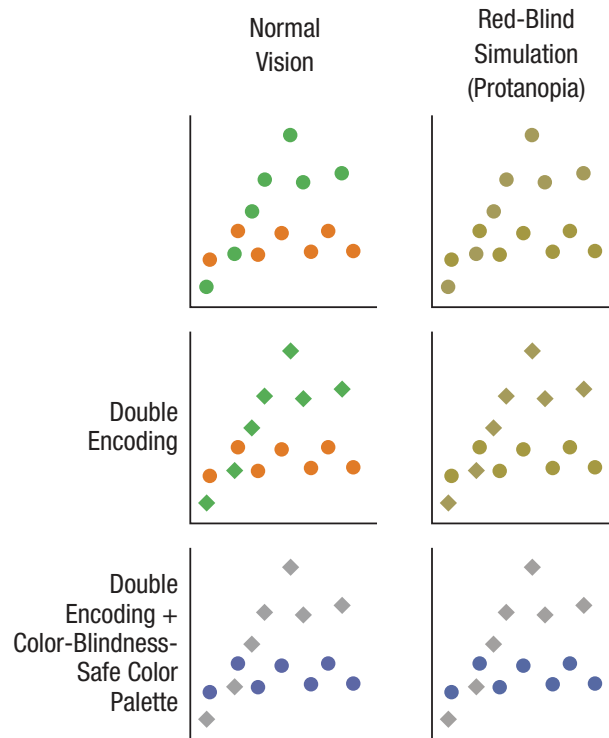


Fig. 5. Three ways to encode data for two groups in a scatterplot, as seen by observers with typical color perception and those with protanopia, a form of color blindness. Protanopia was simulated in Photoshop.

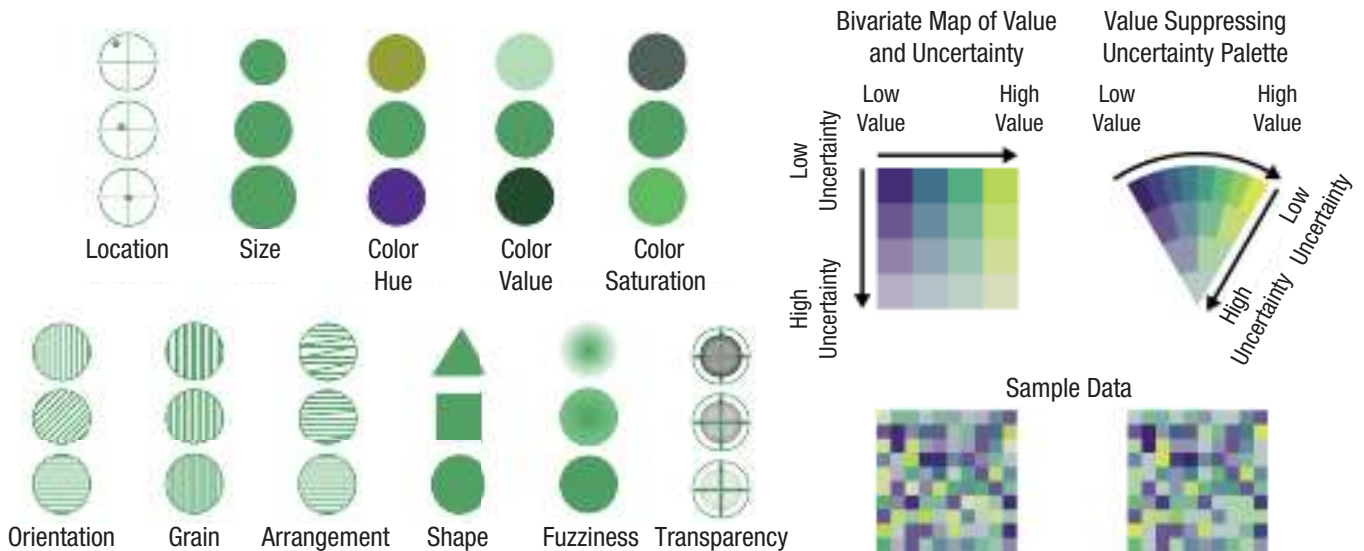


Fig. 23. Methods for depicting uncertainty. On the left of the figure are different visual variables for conveying uncertainty tested by MacEachren et al. (2012), who found that fuzziness, location, arrangement, and color value were rated as relatively more logical than other variables for representing uncertainty. The right shows a bivariate color map that separately encodes value (via hue) and uncertainty (via lightness). A value-suppressing uncertainty palette uses a tree metaphor to collapse value distinctions in proportion to the uncertainty surrounding the values (Correll et al., 2018). Left figure reprinted with permission from “Visual Semiotics & Uncertainty Visualization: An Empirical Study,” by A. M. MacEachren, R. E. Roth, J. O’Brien, B. Li, D. Swingley, and M. Gahegan, 2012, in *IEEE Transactions on Visualization and Computer Graphics*, 18(12), p. 2497 (<https://doi.org/10.1109/TVCG.2012.279>). Copyright 2012 by IEEE. Right figure republished with permission of the Association for Computing Machinery, “Value-Suppressing Uncertainty Palettes,” by M. Correll, D. Moritz, and J. Heer, *CHI ’18: Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (<https://doi.org/10.1145/3173574.3174216>). Copyright 2018 Association for Computing Machinery.

The Science of Visual Data Communication: What Works

Steven L. Franconeri¹, Lace M. Padilla², Priti Shah³,
Jeffrey M. Zacks⁴, and Jessica Hullman⁵

¹Department of Psychology, Northwestern University; ²Department of Cognitive and Information Sciences, University of California, Merced; ³Department of Psychology, University of Michigan; ⁴Department of Psychological & Brain Sciences, Washington University in St. Louis; and ⁵Department of Computer Science, Northwestern University

Abstract

Effectively designed data visualizations allow viewers to use their powerful visual systems to understand patterns in data across science, education, health, and public policy. But ineffectively designed visualizations can cause confusion, misunderstanding, or even distrust—especially among viewers with low graphical literacy. We review research-backed guidelines for creating effective and intuitive visualizations oriented toward communicating data to students, coworkers, and the general public. We describe how the visual system can quickly extract broad statistics from a display, whereas poorly designed displays can lead to misperceptions and illusions. Extracting global statistics is fast, but comparing between subsets of values is slow. Effective graphics avoid taxing working memory, guide attention, and respect familiar conventions. Data visualizations can play a critical role in teaching and communication, provided that designers tailor those visualizations to their audience.

Keywords

visual communication, graph comprehension, reasoning, statistical cognition, uncertainty communication, data visualization

This report presents research-backed guidelines for creating powerful and intuitive visualizations oriented toward communicating data to students, coworkers, and the general public. We begin by reviewing guidelines for helping viewers extract data from visualizations in precise and unbiased ways, avoiding a set of known illusions and distortions. We then describe when visual processing of visualizations is powerful (processing broad statistics) versus where it slows to a crawl (making individual comparisons), and we provide a tool kit for avoiding that slowdown. We review guidelines for ensuring that a viewer properly maps visualized values to the right concepts in the world (e.g., viewers can extract the size of an error bar on a graph, but do they understand what it means?), allowing viewers to use visualizations as effective tools for reasoning. We then review guidelines for conveying uncertainty and risk (e.g., how could a physician express survival odds for a treatment to a patient?). Finally, we summarize a set of guidelines for creating

visualizations that communicate clearly and suggest resources for readers interested in learning more.

Data visualizations range from simple graphs in elementary school classrooms, to depictions of uncertainty in election forecasts in news media, to complex data displays used by scientists and analysts. When designed effectively, these displays leverage the human visual system's massive processing power, allowing rapid foraging through patterns in data and intuitive communication of those patterns to other viewers. But when designed ineffectively, these displays leave critical patterns opaque or leave viewers confused about how to navigate unfamiliar displays.

We review methods, empirical findings, theories, and prescriptions across the fields of visual perception,

Corresponding Author:

Steven L. Franconeri, Department of Psychology, Northwestern University
Email: franconeri@northwestern.edu

graph comprehension, information visualization, data-based reasoning, uncertainty representation, and health risk communication. These research communities study similar questions and use complementary expertise and styles of inquiry, yet they too rarely connect. We ignore artificial boundaries among these research fields, and instead integrate across them.

The Importance of Visualization Design and Literacy

Thinking and communicating with data visualizations is critical for an educated public (Börner et al., 2019). Science education standards require students to use visualizations to understand relationships, to reason about scientific models, and to communicate data to others (National Governors Association Center for Best Practices and Council of Chief State School Officers, 2010; National Research Council, 2013). Evidence-based public policy prescriptions about climate change, vaccines, and policing are argued to be most effectively built (Kohlhammer et al., 2012) and communicated to the public (Otten et al., 2015) with visualizations. Journalists at *The New York Times* Upshot, FiveThirtyEight, *The Economist*, and *The Washington Post* use visualizations to communicate data and evidence about statistics and policy. Data visualizations are ubiquitous in the workplace—in data-analysis software, in data-overview dashboards, and in millions of slide presentations created each day (Berinato, 2016; Parker, 2001). Physicians rely on them to show data about the risks of medical procedures, and meteorologists use them to show the uncertainty in a hurricane's potential path (Ancker et al., 2006; Ruginski et al., 2016).

In each of these domains, low graphical literacy and ineffective design lead many viewers to struggle to understand these otherwise powerful thinking tools. Many students can find textbook visualizations too challenging to understand or integrate with nearby text (Nistal et al., 2009; Shah & Hoeffner, 2002). Public policy visualizations can be counterintuitively designed, leading many viewers to draw a conclusion opposite the one suggested by the depicted data (Engel, 2014). Dozens of best-selling guides have decried the state of visualizations in the workplace and offered prescriptions for more powerful, clear, and persuasive graphs (see the Recommended Practitioner Books section at the end of this article and Ajani et al., 2021, for a more exhaustive list). Medical-risk visualizations can lead patients to fundamentally misunderstand the base rates or risk factors for diseases or medical procedures (Ancker et al., 2006). When a prediction has a high level of uncertainty that is not intuitively conveyed, the public can lose trust in scientists and analysts. For example,

when a hurricane's path deviates somewhat from the most likely trajectory, or when a politician with a 20% predicted chance to win an election prevails, these outcomes may be consistent with the uncertainty inherent to the predictions. But if the forecaster does not effectively visually communicate that uncertainty, their reputation can suffer when their prediction is “wrong” (Padilla et al., 2021).

Who Studies the Design and Comprehension of Visualizations?

Research on the design and pedagogy of data visualizations takes place in several communities. A psychologist focusing on perception might study the mapping between a color value in a heat map and the abstract magnitude that an observer extracts from it (Stevens, 1957). A cognitive psychologist might explore how working memory limits the complexity of the statistical relationships that a viewer might extract (Halford et al., 2007; Padilla et al., 2018). An education researcher might try to remove roadblocks for students struggling to translate visual depictions to their underlying concepts (Börner & Polley, 2014; Shah & Hoeffner, 2002) or seek multimedia design principles for designing effective graphics and integrating them with text (e.g., Mayer & Fiorella, in press). Researchers in public policy communication or political science might study why viewers find some visualizations to be more trustworthy or persuasive than others (Nyhan & Reifler, 2019). Health communication researchers evaluate how to effectively communicate the risk of a medical procedure to patients with low *numeracy* (i.e., ability to work with numbers and mathematics; Ancker et al., 2006). Specialists in statistical cognition and communication seek ways to communicate uncertainty across election outcomes, bus arrival times, and hurricane paths (Hullman, 2019). Finally, a research community housed in computer and information sciences studies data visualization at multiple levels, from data types and algorithms to the creation of user task taxonomies, to design prescriptions for visually powerful displays and fluid interaction (Munzner, 2014).

In this article, we also draw advice from communities of practitioners who might not engage in empirical research but use extensive in-context experience to generate prescriptions for powerful and intuitive visualizations. At the end of this review, we include a list of recommended visualization-design guidebooks. Although many of these guides are oriented toward business analysts, their prescriptions extrapolate directly to science, education, and public policy visualizations. We also discuss design techniques used by a new wave of journalists focused on communicating data analysis to the general public.

The Structure of Our Review

This review focuses on how to effectively design visualizations that communicate data to students and the general public. We review evidence-based prescriptions for designing visualizations that help people understand and reason about the patterns, models, and uncertainties carried by a data set. Another important topic, which we do not cover systematically here, is how to measure visualization literacy and the effectiveness of teaching techniques that improve it (see Börner et al., 2019; Lee et al., 2016). We also restrict our scope to quantitative visualizations, omitting discussion of qualitative visualizations of text data, diagrams, and processes (see Hegarty, 2011; Henderson & Segal, 2013, for review). We focus on research and prescriptions that are most relevant for communication to nonspecialist audiences, instead of the design of powerful tools for data analysis within expert communities.

We first illustrate why visualizations can be such powerful tools for thinking about data. Because the human visual system is highly developed for rapid parallel extraction of behavior-relevant features and relationships, visualizations allow us to process some types of patterns across an entire two-dimensional array of values at once. We describe the limited set of visual channels that can effectively depict magnitudes to a viewer, such as the *position* of a value in a dot plot, the *size* of a circle hovering over a map, or the *color intensity* of an activation pattern in a functional MRI (fMRI) image.

We then discuss design guidelines for ensuring that the human eye accurately decodes those depicted values. We review evidence for a ranking of some visual channels (e.g., position) as more precise than others (e.g., color intensity) for at least one common task but also discuss how new work has begun to dismantle that ranking for a broader array of tasks. We list a set of common errors and illusions that cause viewers to extract the underlying values from visual channels incorrectly—for example, *y*-axis manipulations that exaggerate differences among values, confusion about whether circles depict values with their size or diameter (which can change the extracted value by an order of magnitude), a common illusion produced by line graphs, and other illusions and categorical distortions that can arise when depicting value with color intensity. We also include a brief review of accessibility considerations for viewers with color blindness. Finally, we discuss best practices for distinguishing between groups of data (say, two groups of points on a scatterplot) by marking them with different shapes or colors.

Next, we discuss an important dissociation in visual processing power: Whereas computing statistics across an image is broad and instantaneous, making comparisons among subsets of values is slow and limited to two or three comparisons per second. We review the

types of grouping cues that loosely control what information is compared by a typical viewer and further techniques for precisely guiding a viewer to the right comparison. We discuss the importance of respecting a viewer's limited working memory, including avoiding legends and animated displays that can engage but also confuse. Finally, we review evaluations of whether visualizations should have rich and memorable designs, as opposed to a minimalist and clean aesthetic.

The next section introduces visualization *schemas*, or knowledge structures that include default expectations, rules, and associations that a viewer uses to extract conceptual information from a data visualization. We illustrate the importance of schemas by introducing the reader to a small set of new visualization designs that will likely be unfamiliar. We then provide examples of common schema elements that are known to more graphically literate audiences (but not always respected by designers), such as the assumption that larger values are plotted upward. We shift to a brief review of human reasoning about visualizations, including formal models that draw links from visual depictions, to numeric values, to their underlying concepts and the designer's intended message. We then explore two case studies: reasoning about graphs illustrating scientific concepts and reasoning about graphs of mathematical functions.

The subsequent sections review research on visualizing uncertainty or risk. Communication failures can start with a lack of understanding of critical statistical concepts, even among scientists. We give examples of how viewers tend to misread error bars as depicting the edges of a range of data instead of correctly understanding them as parameters of a distribution. Probability information expressed as risk is critical for people such as patients considering a medical procedure and potential evacuees who may be in the path of a hurricane, but depictions of risk are frequently misunderstood. We review guidelines for showing uncertainty or risk more intuitively, including depicting samples of discrete outcomes, showing probability density functions, and depicting data with arrays of icons.

Finally, we summarize a set of evidence-based prescriptions for creating powerful visualizations for intuitive communication of data and provide a list of recommended practitioner guides (Box 1), websites, and data-journalism outlets for further reading and inspiration (for a concise review of similar guidelines, see also Zacks & Franconeri, 2020).

The Power of Visualization

Visualizations let viewers see beyond summary statistics

Visualizations allow powerful processing of an entire two-dimensional rectangle of information at once, in

Box 1. Recommended Reading

The following books offer excellent, concise guides to designing effective visualizations for communication of data and analyses:

Camões, J. (2016). *Data at work: Best practices for creating effective charts and information graphics in Microsoft Excel*. New Riders.

Evergreen, S. D. H. (2017). *Presenting data effectively: Communicating your findings for maximum impact* (2nd ed.). SAGE Publications.

Knafllic, C. N. (2015). *Storytelling with data: A data visualization guide for business professionals*. Wiley.

Schwabish, J. (2021). *Better data visualizations: A guide for scholars, researchers, and wonks*. Columbia University Press.

For a review of the ethics of persuasion with visualizations, we recommend the following books:

Cairo, A. (2016). *The truthful art: Data, charts, and maps for communication*. New Riders.

Cairo, A. (2019). *How charts lie: Getting smarter about visual information*. W.W. Norton.

For a review of accessibility considerations in data visualization, we recommend the following article:

Kim, N. W., Joyner, S. C., Riegelhuth, A., & Kim, Y. (2021). Accessible visualization: Design space, opportunities, and challenges. *Computer Graphics Forum*, 40(3), 173–188.

stark contrast to the limitation of reading handfuls of symbolic numbers per second. As a demonstration, Figure 1 (top left) contains four sets of 11 pairs of values. Take a moment to compare those columns, and notice that reading symbolically represented numbers takes time. As you seek patterns within each set, or make comparisons among the four sets, progressively processing more pairs of values becomes increasingly difficult. Worse, these tasks quickly exhaust memory capacity, such that new numbers or patterns tend to displace ones that were previously seen. These limitations on symbolic processing of numbers lead viewers to rely instead on summary statistics that compress data sets into a single group of numbers. For the four sets of numbers in Figure 1, those statistics on the bottom row—means, standard deviations, and correlation coefficients—are identical, which might lead you to believe that the numbers contributing to the statistics are similar (Anscombe, 1973). However, because statistics summarize larger sets of numbers by abstracting over them and making assumptions about the patterns that they might contain, many sets of numbers can generate the

same statistics. For these four sets of numbers, relying on statistics turns out to be dangerous.

The differences between the four sets of numbers in Figure 1 are immediately visible when translated into images in the form of scatterplots (bottom left). These images allow you to leverage the two-dimensional processing power of your visual system, the largest single processing system of the brain (Van Essen et al., 1992). If you are familiar with statistics, then the first image at the bottom left likely matches what you assumed the numbers should look like given the statistics on the bottom rows: an orderly positive relationship. But the other sets are clearly different in important ways. The right side of Figure 1 depicts frames of an animation that provides a more sophisticated example (Matejka & Fitzmaurice, 2017), in which visual processing allows one to immediately see that, despite identical statistics (to the second decimal place) for each scatterplot, the nine plots contain saliently different patterns. When exploring a new data set, it is good practice to visualize every column of data with a histogram, and every potentially interesting pairing of columns with scatterplots, before turning to statistical summaries (Moore et al., 2017; Wongsuphasawat et al., 2015).

Visual channels translate numbers into images

Visualizations rely on several visual channels to transform numbers into images that the visual system can efficiently process (Bertin, 1983; Mackinlay, 1986; see Munzner, 2014, for a more complete list). Knowing these channels allows a designer to consider which might be best suited for a given data set and context—particularly given that each is associated with differential levels of precision and potential illusions, as described in the following sections. The first column of Figure 2 depicts five of the more frequently used channels. Dot plots and scatterplots, such as those in Figure 1, represent values as *position*. Bar graphs represent values not only as positions (of the tips of the bars) but also as one-dimensional *lengths* (and, some argue, even two-dimensional *areas*; Yuan et al., 2019). If two bars do not rest on the same baseline, such as segments within the same bar in a stacked bar graph, the comparison relies only on length or area. Next, two circles code numbers exclusively as two-dimensional areas (typically circles), a technique often used to overlay values across maps. *Angle* typically emerges when points are connected to form a line graph, organically allowing an encoding of the difference between adjacent points (a bigger difference creates a steeper slope and, typically, a longer line). Outside of pie charts, angle is less frequently used to

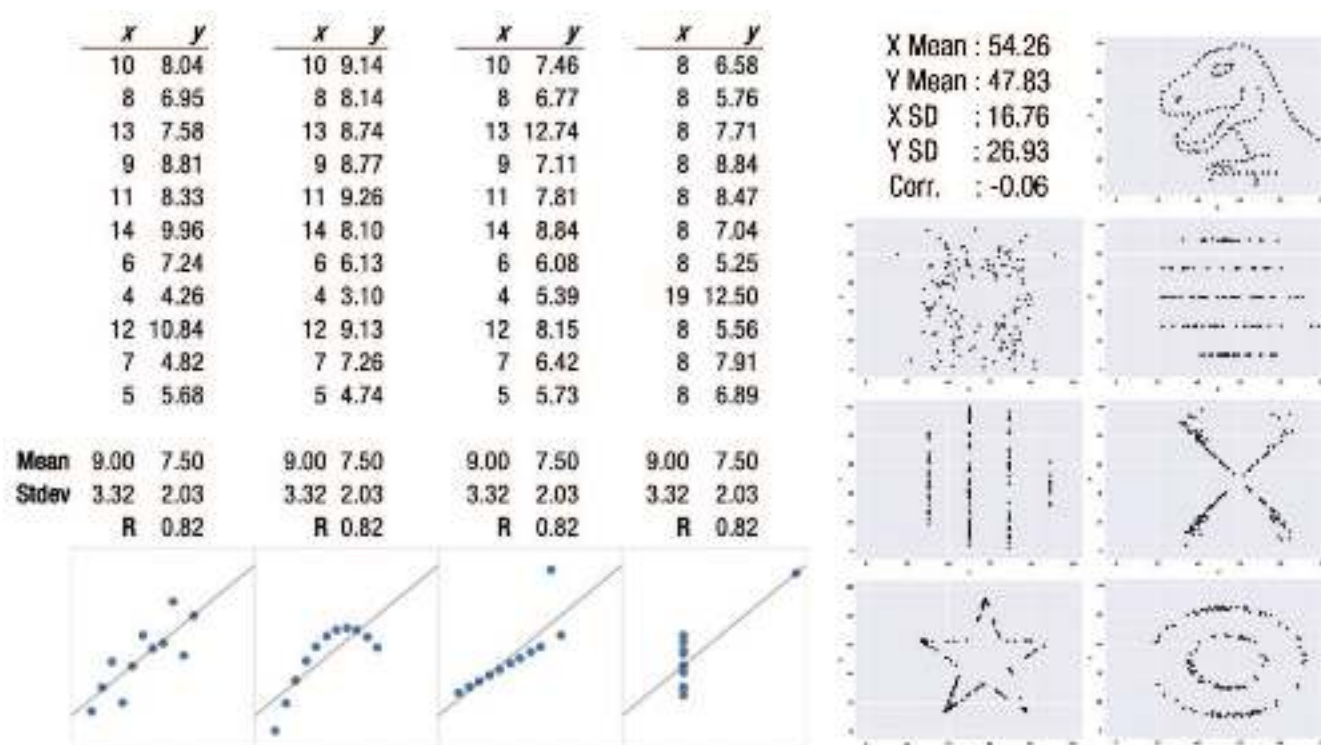


Fig. 1. Examples of how visualizations can let viewers see beyond summary statistics. At left, four sets of 11 numbers have identical statistics but dramatically different patterns, as revealed by the scatterplots below each column. At right is a more extreme example of nine dramatically different scatterplots (including one that looks suspiciously like a dinosaur) depicting data with identical statistics, down to the second decimal place. The graphs on the right are adapted with permission of the Association for Computing Machinery, from “Same Stats, Different Graphs: Generating Datasets With Varied Appearance and Identical Statistics Through Simulated Annealing,” by J. Matejka and G. Fitzmaurice, *CHI '17: Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (<https://doi.org/10.1145/3025453.3025912>). Copyright 2017 Association for Computing Machinery.

depict numbers directly—perhaps on local areas of a map to represent wind directions. *Numerosity* is omitted from the figure, but it often implicitly shows higher-level attributes of data. For example, you can immediately estimate the number of points in a scatterplot, segments in a stacked bar chart, or icons in an infographic. *Intensity* is an umbrella term (often also called *lightness* or *value*) for either luminance contrast or color saturation, as used in a heat map or fMRI activation map. *Motion* is also not included in the figure, but animating a scatterplot to show values changing over time can encode the rate and direction of change in the speed and direction of the dots' motion.

How to Design a Perceptually Accurate Visualization

Understand how to leverage visual channels

Visual channels are ranked by their perceptual accuracy. These channels differ in how precisely they convey numeric values to a viewer, and knowing the

ranking of these channels allows a designer to prioritize what information to show most precisely. The leftmost column of Figure 2 presents five of the channels that can depict metric data to the human visual system. This list is ordered by the typical precision with which a viewer can verbally state the ratios between the two values shown; more precise ways of communicating numbers are at the top and less precise ways are at the bottom (Cleveland & McGill, 1984, 1985; Heer & Bostock, 2010). It should be clear from the figure that the 1:7 ratio can be relatively precisely extracted for position, but that task is a bit tougher for area, and far more difficult for intensity, at the bottom of the list.

Because position is the clear winner for precision, visualization designers often prioritize the vertical and horizontal dimensions of two-dimensional space when depicting or organizing quantitative data. Faced with a single column of numbers in a spreadsheet, a visualization designer might depict those data vertically with position (in a bar or line graph) and rely on horizontal position to organize the values into categories, as in a typical bar chart. If faced with two columns of numbers, a designer might simply create two of those same types

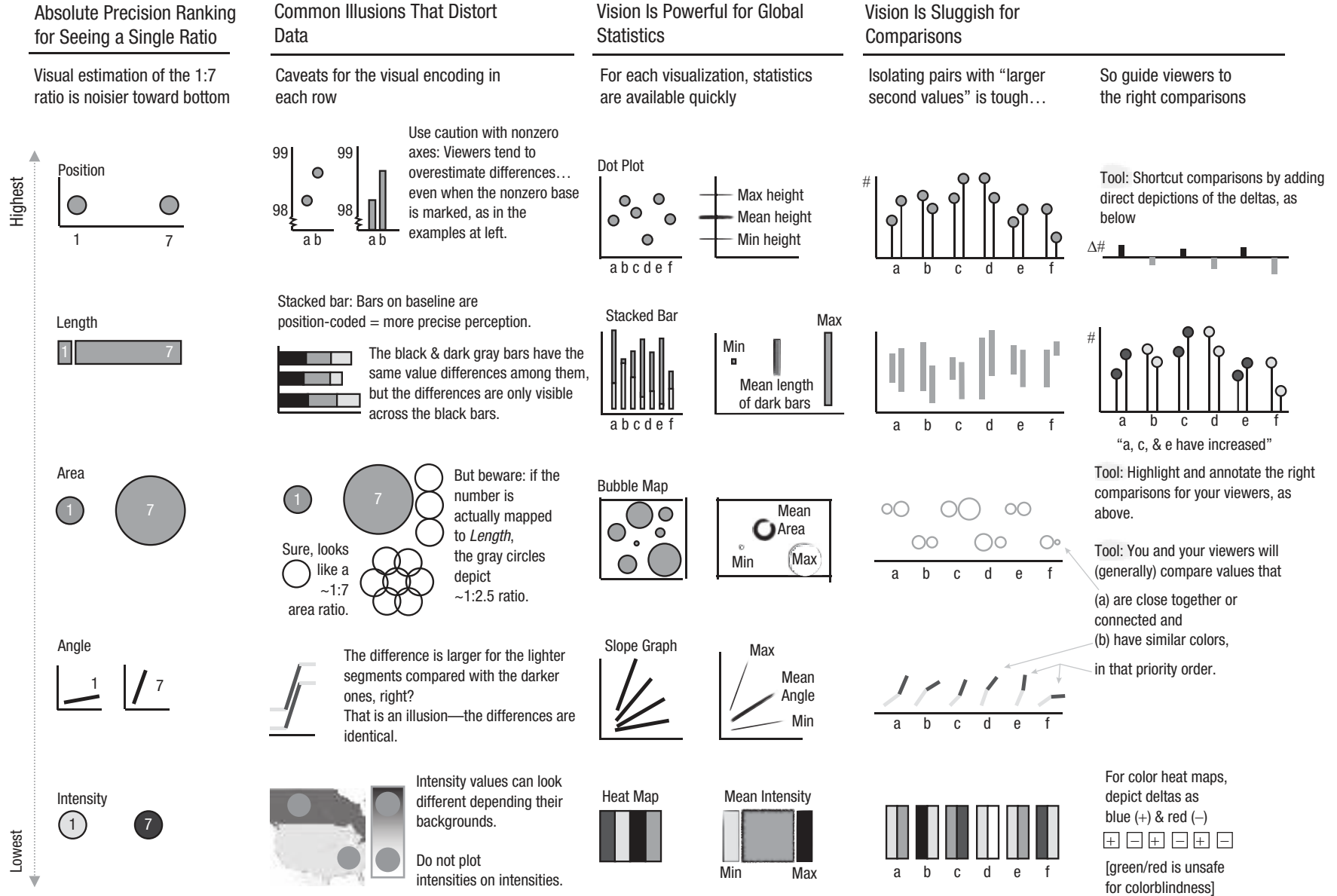


Fig. 2. A summary of the power and limits of visual data processing. The two columns on the left show a quick reference guide to channels that can depict data visually and common illusions for each channel. The column in the center presents a summary of how visual statistics are powerful. The two columns on the right illustrate how comparisons are severely limited and present a set of design techniques that focus viewers on the "right" ones.

of graphs or organize each set of numbers along both the vertical and horizontal axes of position, as in a scatterplot.

The advantage of position over length for precisely depicting ratios between numbers is demonstrated in the second column of Figure 2, which shows a horizontally oriented stacked bar graph in its second row of examples. Because the black segments of the bars are aligned on a common axis at their left, their right tips provide a precise position code, allowing the viewer to see the delicate 0.9:1 ratio between the bars. However, the next set of medium-gray segments are tougher to distinguish because the positions of their right tips are no longer useful, so the viewer must rely on length—a lower-precision channel—to extract the same ratio.

Mapping visual ratios back to numbers can cause perceptual errors.

Using visualizations can unlock powerful data-pattern processing. However, a designer must be aware of several perceptual illusions that can lead viewers to map visual depictions back to their original numeric values incorrectly (Huff, 1954; Tufte, 1983). If two plotted values have a 1:7 ratio, then the visualization should cause a typical viewer to see that 1:7 ratio veridically. Even for a precise visual channel such as position, this requirement can be tougher than anticipated. For example, see the dot plot and bar graph at the top of the second column of Figure 2. The dot plot uses position as its visual channel, and the bar graph depicts the same data with both position and length. The second value appears to be roughly double the first value. Look more closely at the y -axis: The second value is only about 1% bigger than the first; the difference appears greater because the axis baseline does not start at zero. In theory, the data are transparently depicted—but in reality, such graphs are frequently misinterpreted.

Figure 3 illustrates some real-world examples of this problem. The line graphs in the upper left, adapted from Darrell Huff's classic 1954 book *How to Lie With Statistics*, show how a line graph's scale can be stretched to make a trend appear steeper (Huff, 1954). In March 2014, a version of the bar graph at the upper right appeared on *Fox News*, a network with an avowedly opposite political orientation to Barack Obama, the U.S. president at the time. Around 6 million U.S. citizens had signed up for a new health-care program sponsored by the president, and the government specified a goal of 7 million sign-ups by March 31. Although the numbers presented are honest (a 6:7 ratio), the visualization's truncation of the y -axis tells a different story (a 1:3 ratio) to the viewer's visual system, suggesting a failure of the president's plan.

Will people not simply read the y -axis labels and easily overcome this initial misperception? Unfortunately, both real-world anecdotes and laboratory studies

suggest that this practice can be deeply misleading (Correll et al., 2020; Hofman et al., 2020; Pandey et al., 2014). For example, using a visualization similar to the *Fox News* example, researchers asked crowdsourced workers to rate the contrast between the two depicted values on a 1-to-5 scale. Ratings in a zero-baseline condition averaged around 1.5, whereas ratings in a deceptive-baseline condition averaged 2.8 (Pandey et al., 2015). In that study, stretching the y -axis also strongly affected ratings of the strength of the trend. The crowdsourced workers were not simply inattentive: Only participants who passed attention-check trials were included. Moreover, the deceptive effect persisted when participants were asked to type the numeric values represented by each bar before making their effect-size rating and were reminded of the y -axis's truncation, which was indicated by a “broken axis” symbol (such as that shown in Fig. 2, top of second column) at the base of the y -axis (Correll et al., 2020).

The prevalence of this deceptive effect has led to quantitative prescriptions for how to set y -axis boundaries to produce accurate measures of statistical effect sizes by typical viewers (Witt, 2019; B. W. Yang et al., 2021). For example, if the relevant data are far from zero, starting the y -axis at zero can make effect sizes illegible. One approach to increase legibility is to center the y -axis on the data's mean, then extend the y -axis 0.75 SD above and below the mean (Witt, 2019). This approach can be appropriate when the overall scale is not essential.

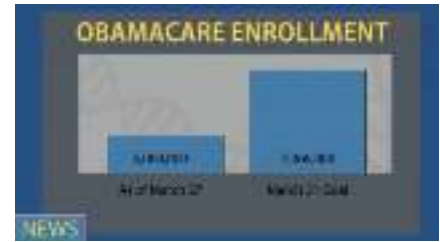
Visually conveying an appropriate difference often depends on what “appropriate” means. Many visualization designers subscribe to the principle that a y -axis must always start at zero so that the visually depicted ratios match the ratios in the data. By contrast, others argue that this guideline must be subject to context (for a distillation of arguments by designers, see Correll et al., 2020). For example, it seems clear that designers should consider the practically relevant range of data values when small but essential differences would be tough to see on a zero-baseline graph. A now-infamous (and now-deleted) story in the *National Review* used a graph similar to one at the bottom left of Figure 3 to suggest that the recent global temperature rise is inconsequential (Correll et al., 2020). It even went beyond using a zero baseline for the axis, intriguingly using -10°F as a lower bound. The graphic to its right shows the pattern of data widely agreed by the scientific community to be more honest: a “hockey stick” pattern indicating a recent rapid rise.

The image at the bottom right of Figure 3 is a now-famous cover of *The Economist* magazine from September 2019, which similarly mapped a restricted range of data to the full range of a blue-to-red color scale. Another issue highlighted by these examples of

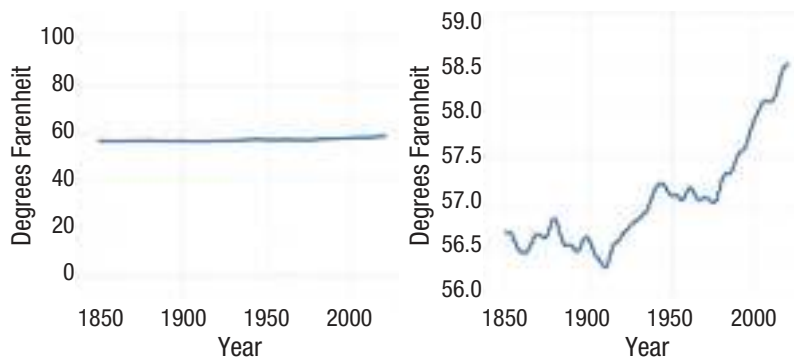
The Difficulty of Mapping Numbers to Visual Channels in Honest Ways



Stretching the y-axis scale of the left graph drastically increases the slope of the perceived trend at right, which feels dishonest.



The same axis stretch in a bar graph, from a real-world example.



Here, taking a small visual increase (left) and stretching it (right) is the “honest” way to show climate-change data.



This color mapping similarly amplifies a relatively small temperature increase.

Fig. 3. Deceptive axis manipulations across a line graph (top left) and a bar graph (top right). So, should data always be plotted relative to zero? The graph on the bottom left depicts climate change by plotting temperature data from a baseline of 0 °F, yet most would agree that graph is less informative than the version to its right, which allows the differences in temperatures to be seen. At the bottom right of the figure, a cover of *The Economist* from September 2019 maps the same restricted range of data to the full range of a blue-to-red color scale. The graphs at top left are inspired by Huff (1954); those at bottom left are inspired by Correll et al. (2020). The magazine cover at bottom right is reprinted, with permission, from *The Economist* (September 19, 2019).

temperature data is that not all variables have a single natural zero point. Zero degrees Fahrenheit is not a meaningful value for ratio comparisons. Temperature in Celsius has a somewhat meaningful zero point (the typical freezing point of water on earth), but that baseline is not relevant to climate change.

These mismappings of visually depicted ratios are not isolated to the position and length channels. One can imagine similar differences in interpretations arising for size or intensity. Imagine a map of a country with a circle over each major city. One might depict crime rates with the intensity of the circles' color, but with what link to the original data? If zero were mapped to gray and 100 violent crimes per capita per year (a very

high rate) to red, the country might seem safe regardless of the underlying data. Another depiction might link 100 violent crimes per year to a bright red. Side by side, those two depictions would suggest a huge difference in crime rates despite showing the same data. One could imagine the same trick, but this time mapping crime rates to the circles' size. If 100 violent crimes were linked to a 1 mm circle, the country would seem safe, but if the same number of crimes were linked to a 10 mm circle, it would seem more dangerous.

Which visual channel is linked to the data values? In other cases, viewers can misunderstand which

visual property encodes the data. In a classic example at the top left of Figure 4, values are encoded by one-dimensional length (the height of each person), producing a 1:2 ratio of the two numbers. However, you might find that your estimate of the depicted values is determined instead by the area taken up by each person, leading to something closer to a 1:4 ratio (or even a 1:16 ratio, if the icons suggest three-dimensional volume). People do indeed make this error, even when the numeric data are printed saliently near the visual representation (Pandey et al., 2015).

In the donut plot to the right of the human icons in Figure 4, there are two ways that numbers are potentially miscommunicated. First, although the data are mapped onto angle, viewer judgments are more strongly determined by two-dimensional area (with some potential contribution from arc length, which is tough to dissociate from area). Because the two donuts are different sizes, this means that the larger donut's value will be artificially inflated. Second, the graph is depicted in 3D. If the viewer can recover the actual three-dimensional geometry from the two-dimensional depiction, then the values should be accurately perceived. By contrast, if the viewer pulls values from the two-dimensional image (the amount of green or purple pixels on the screen), the values in the "front" will be inflated because of the perspective projection. Unfortunately, this technique is indeed substantially misleading because static two-dimensional projections do not typically lead to effective recovery of three-dimensional structures (Tittle et al., 2001; but see Brath, 2014).

Avoid common illusions and misperceptions

A common optical illusion in line graphs. In the line graph in the middle of Figure 4, the two curves are identical ($y = x^3$), but the darker line is translated vertically upward by a constant of 1,000. Even if the viewer knows that the two shapes are identical, it is difficult to see that the vertical distance between the two lines is the same across their entire horizontal span. Instead, given any point on the dark line, viewers tend to see its distance from the closest point on the gray line, which becomes progressively smaller as both lines increase. This illusion makes it difficult to visually estimate differences between lines, especially lines with steep slopes (Cleveland & McGill, 1984). A similar example is depicted in the second column of Figure 2. This illusion is well known to electrophysiology researchers: When faced with visualizing the difference between two measured waves, they will explicitly plot a "difference wave" that shows the difference as a single line (Luck & Kappenman, 2012).

Illusory contrast effects for intensity. The bottom of the second column of Figure 2 shows a final illusion that

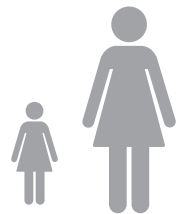
can warp our perception of visualized data. Both on the map and in the rectangle, the two vertically separated circles have the same luminance value. However, the lower circle is subjectively darker to the eye because it has been placed on a lighter background and has a higher contrast with its surroundings. In the real world, converting luminance to contrast is a critically important mechanism for seeing accurate luminance and color despite changes in the brightness and color profile of light in the environment (Purves et al., 2004). However, this correction leads to misperceptions of intensity-coded values in the artificial world of data visualizations (Szafir, 2018). One rule of thumb is never to plot intensities on top of other intensities that vary, as in the map in Figure 2.

Misleading illusions that combine separate values. When plotting two sets of numbers on a map, designers typically rely on intensity for one set of values and map the other set of values to the area of circles. This design solution works because, rather than plotting one intensity on top of another intensity, which creates an *integral* representation of contrast, intensity and area are relatively separable representations (Garner, 1974). Other examples of integral representations include encoding two sets of data in rectangles—one set in their widths and one set in their heights. Instead of seeing these values separately, the eye is tempted to translate them into the aspect ratio and the area of each rectangle. The eye then focuses on the ratios and multiplication of each pair of values (Shechter & Hochstein, 1992). As an extreme case, it is unwise to attempt to use the red component of a single color (imagine using the RGB sliders to change the color of an object in presentation software) to depict one number and the green component for another. Red and green will combine in an integral fashion when both are at their highest value, and the viewer will see a single integral percept of yellow (Ware, 2019).

A final example is shown in the scatterplot in Figure 4. Once two sets of numbers are combined into a single two-dimensional plot, new integral percepts emerge, such as the distance between any two points across both their x and y values, points that are outliers on both axes, or the global shape of all points that we can easily interpret as a correlation (F. Yang et al., 2019). However, there is a trade-off. The distribution of values of either set in isolation is now tougher to disentangle (Mackinlay, 1986). This is why data scientists often pair a scatterplot with "marginal histograms" that allow them to see those data in a separable way.

The biasing effect of categorical perception. When continuous values are encoded through visual channels, those values can be warped by categorical perception. A classic example is the seven discrete colors that we see in a rainbow, which are not present in the rainbow itself.

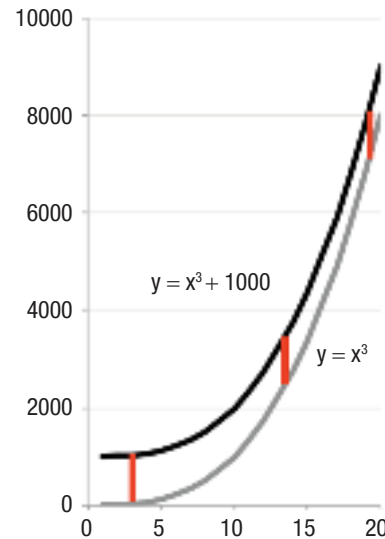
A Set of Common Visual Confusions, Illusions, and Distortions



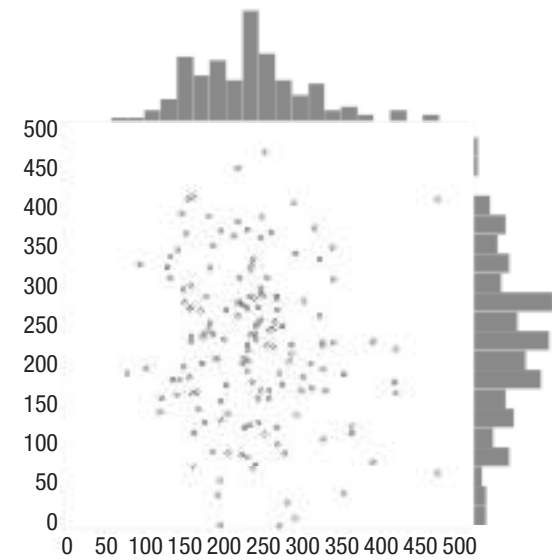
In these examples, linear increases in height or radius suggest the square of that increase.



The same data are displayed with two color mappings, revealing how color-category boundaries can bias value perception.



The difference between these lines is constant, yet an illusion suggests that the difference decreases.



Combining two data dimensions in a scatterplot can make recovery tougher for either alone.

Fig. 4. A set of common visual confusions, illusions, and distortions. At upper left, the two human icons represent vastly different ratios depending on whether the data are represented by their one-dimensional heights, their two-dimensional areas, or their three-dimensional volumes. The donut graph to their right shows that three-dimensional depictions can artificially inflate data values in the two-dimensional plane. At bottom left, the same fMRI data are plotted with two color mappings. The brain image on the left produces less of a spurious categorical boundary effect, whereas the image on the right shows a common red-to-green color map that makes continuous variation appear exaggerated when it maps onto transitions from one color category to another. At center, an illusion prohibits the accurate recovery of differences between values in a line graph. At right, correlation is easier to detect from the scatterplot, but individual histograms for each dimension are easier to see when plotted separately.

Those color categories are created by an automatic process that systematically bins continuous wavelengths into one of several perceptual categories, exaggerating metric differences among values that straddle those boundaries and shifting percepts toward prototypes for the categories (Goldstone & Hendrickson, 2010; Newcombe et al., 1999). This same phenomenon occurs when data are depicted by hues, as in the bottom left of Figure 4. Across the two brain images, the one on the right uses a greater variety of hues to depict activation values. These additional hues create new color-category boundaries that can dramatically exaggerate the differences between values that straddle them (Y. Liu & Heer, 2018; Quinan et al., 2019). In *The Economist* cover shown in Figure 3, the blue-to-red scale creates a salient categorical color boundary at the color transition point, which makes the temperature increase in the past few years especially salient. Similar category boundaries can affect the perception of values depicted by other channels such as position or length. In a pie graph or stacked bar graph, values that are near gridlines or the implicit 50% mark in the middle of the bar or pie are recalled as being farther from that category boundary (Ceja et al., 2021; McColeman et al., 2021; Spence & Krizel, 1994; Xiong, Ceja, et al., 2020).

Design for color-vision impairments

Beyond illusions and biases, certain combinations of colors in visualizations can be problematic for people who are color-blind or have other color-vision impairments. Color-vision impairments are estimated to affect 4% of the global population (Olson & Brewer, 1997), or roughly 300 million people. Further, older adults can have less sensitivity to color (Silva et al., 2011). Some color blindness results in viewers not being able to distinguish between various colors; protanopia, or red-green color blindness, is the most common, but various other versions of color blindness exist. Color-vision impairments are highly problematic for visualizations, as a portion of the audience may literally not see important patterns in data.

Numerous online color-blindness simulators allow users to upload an image to learn how someone with color blindness would see it (Asada, 2019). For example, the first row in Figure 5 shows a scatterplot encoded with two colors, green and orange. People with typical vision can see that the green dots have a steep positive correlation and the orange dots make a flat line. However, when the scatterplot is processed through a color-blindness simulator, the colors look the same, and all the dots appear to show a positive correlation. The simplest way to make visualizations accessible to viewers with color blindness is to avoid using hue as the only encoding channel or allow viewers to change the

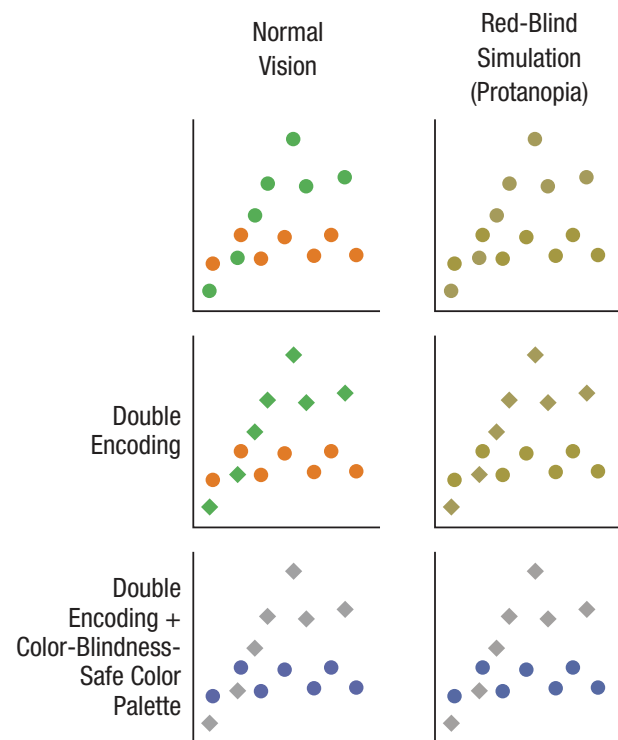


Fig. 5. Three ways to encode data for two groups in a scatterplot, as seen by observers with typical color perception and those with protanopia, a form of color blindness. Protanopia was simulated in Photoshop.

color palette (Silva et al., 2011). Designers can also double-encode a variable, using hue and another encoding channel (Plaisant, 2005), as in the second row of Figure 5. The most thorough and inclusive option is to use color palettes that are safe for people with color blindness, such as those proposed by Harrower and Brewer (2003), as seen in the bottom row of Figure 5.

Design for perceptual accuracy across a broad array of tasks

The ranking of precision for visual channels depicted in the first column of Figure 2 is based on a measure of precision in a particular task: the average error in verbal reports of the ratio between two depicted values (1:7 for each of the examples in the first column; Cleveland & McGill, 1984; Heer & Bostock, 2010).

Although this task is surely important when comparing two values, it is not the only task (or perhaps even the primary task) that viewers complete when examining visualizations (Bertini et al., 2020). One proposed taxonomy of graph-interpretation tasks separated them into three levels: elementary, intermediate, and “overall,” roughly corresponding to simple fact retrieval, comparison and identification of trends, and gist understanding (Bertin, 1981). Additional taxonomies have been developed in the context of cognitive models of

graph comprehension. Other taxonomies are based on analyses of cognitive processes involved in different tasks (Tan & Benbasat, 1990), reliance on local versus global features (Carswell, 1992), or other criteria (MacDonald-Ross, 1977; Washburne, 1927; Wickens, 1989). Work in the data-visualization literature has begun to catalog such tasks and to test the types of displays that best support performance for each. One list of tasks is distilled from questions submitted by students who were asked to analyze a large data set (Amar et al., 2005). This list includes analytic tasks such as “retrieve value,” “sort,” “determine range,” “correlate,” and “characterize distribution.” Another list of tasks focuses on how people generate summary statistics for plotted data (Szafir et al., 2016).

The ranking derived from performance on two-value ratio judgments does not always extrapolate across these alternative tasks. For example, depicting data with a line graph, which relies on “precise” position coding, can lead to *lower* efficiency in seeing big-picture statistical properties such as means. Intriguingly, the opposite is true for intensity coding. Multiple studies have shown that for identifying particular values, position is far more precise than intensity, but for judging an average across many values, intensity is more precise (Albers et al., 2014; Correll et al., 2012). The reason for this dissociation is not well understood. It is possible that the lower precision of intensity better allows its values to blend together to construct aggregate values, or perhaps intensity is simply processed by a different mechanism that affords aggregate judgments (Szafir et al., 2016). Some visualization designs purposely use the low precision of the intensity channel to focus viewers on the big-picture trend of the data instead being distracted by precise details, as in the visualization from the cover of *The Economist* magazine (Fig. 3), which focused viewers on the big picture of climate while de-emphasizing more detailed (but less relevant) variability in monthly weather.

Other work has tested whether people can more quickly or accurately complete the types of tasks listed above (judge mean, correlate, etc.) given different graph formats (scatterplots, bar graphs, tables, etc.) and data-set sizes (small vs. large; Y. Kim & Heer, 2018) or different arrangements of data within a graph (Jardine et al., 2019; Ondov et al., 2019). Although such studies have found consistent interactions—better performance on Task X with Design Y in Arrangement Z for Data-Set Size S—the existing pattern of results is currently too complex to generalize to novel combinations. To create generalizable guidelines, the research community will likely require a more complex model of the underlying perceptual operations that produce these complex interactions (Jardine et al., 2019; Ondov et al., 2021).

Among these alternative tasks, the perception of correlation is particularly well studied (Harrison et al.,

2014; Jardine et al., 2019; Rensink & Baldridge, 2010; F. Yang et al., 2019; Yuan et al., 2019), leading to initial suggestions of its underlying perceptual operations. For example, scatterplots allow for more accurate judgments of correlation than pairs of bar charts presenting the same data (Harrison et al., 2014).

Effectively distinguish among groups in data

Scatterplots present two metric variables in relation to each other. Imagine that the horizontal position represents the longitude of a collection of oil wells and the horizontal position represents their latitude. This example makes clear that scatterplots and data maps share DNA. Suppose the designer now wants to differentiate the set of points according to a *nominal* (categorical) variable, such as the company that owns the oil well. Because the two dimensions of position (vertical and horizontal) are already being used to represent the metric variables, designers would typically differentiate points from each group with either different colors or shapes.

Plotting the groups as categorically different colors is often the first choice because the visual system processes color differences more efficiently than shape differences across the two-dimensional visual plane, as measured by performance in visual-search and texture-segmentation tasks (Wolfe & Horowitz, 2017). These findings, based on simple displays used in laboratory studies, extrapolated to a data-visualization context in which viewers compared the average heights of multiple color-coded or shape-coded clouds of points in a scatterplot: Color coding produced far better performance (Correll et al., 2012).

When color differences distinguish data from two separate groups or classes, differentiating those classes is easier if the encoded colors are farther apart in a perceptual color space. For example, it is easier to differentiate red from blue than from orange-red. Researchers have constructed effective palettes from perceptually informed color spaces (e.g., CIELAB; Y. Liu & Heer, 2018) to suggest colors to use to differentiate N nominal classes; colors become progressively crowded together as N increases. One such set that is viewable (and customizable) online is ColorBrewer 2.0 (Brewer, 1994a, 1994b). Another tool, Colorgorical, balances perceptual differentiation with aesthetic considerations (Gramazio et al., 2016).

Picking a color for a nominal value should also be constrained by the semantic congruence of the value and the color. If the nominal values are lemons and cherries, it is easier for viewers to answer questions about a chart that labels those values with yellow and red, compared with a standard palette that does not consider semantic congruence (Lin et al., 2013; Schloss

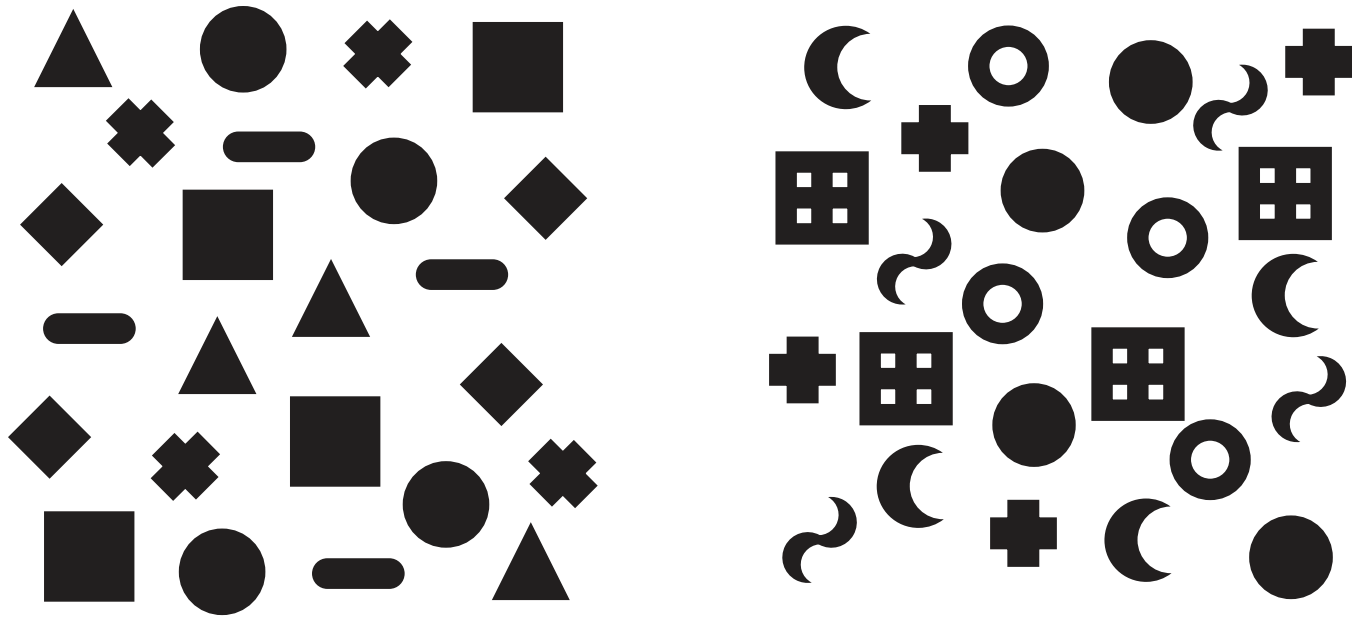


Fig. 6. The standard shape set for Microsoft Excel (left) compared with a perceptually spaced set (right; inspired by Huang, 2020). Try to pick out the four instances of each shape in each display—you should find that task easier on the right side.

et al., 2018). An algorithm can automatically generate intuitive color choices for a given noun by analyzing the color profile of images pulled from Web searches for the noun and then optimizing color assignments in terms of perceptual spacing and semantic fit. Such algorithms can perform as well as human experts in quickly picking color palettes for nominal data (Lin et al., 2013; Setlur & Stone, 2015).

Sometimes a visualization designer needs to show a second set of nominal values in the same plot. Thinking back to our map, imagine that we have used a unique color to represent each oil company (A, B, C) but also wish to depict each company's nationality (Canada, Brazil, Mexico). Typically, shape would be used to show that second nominal variable. Because it is less perceptually effective than color, it should be used for the less important variable, or the one with fewer values to differentiate. The shape sets used in commercial software (e.g., Microsoft Excel) gravitate toward intuitive shapes, such as circles, triangles, squares, and diamonds, that are not actually well separated in perceptual space.

Researchers have begun to explore perceptual shape space, and some have extrapolated from pairwise subjective similarity ratings of combinations of candidate shapes (Demiralp et al., 2014). Other work has relied on objective performance tasks on actual simulated scatterplots (Burlinson et al., 2017; Huang, 2020). This work is also new, but so far, human shape space (at least for the simple shapes used in visualizations, and at least for the types of tasks tested so far) appears to prioritize the difference between open (circle, square, triangle) and closed ($\times$, $+$, $*$) shapes, such that

differentiating points is easier when they differ in that property (Burlinson et al., 2017; Huang, 2020). An initial full three-dimensional perceptual shape space (Huang, 2020) adds the additional properties of intersection and spikiness; Figure 6 depicts a clear improvement in shape differentiability compared with the typical sets used even in professional data-visualization software.

Finally, some software automatically differentiates nominal variables with both color and shape, under the assumption that more differentiation is better. However, work has shown that color is already so dominant in its effectiveness that *redundant encoding* does not substantially improve visual processing efficiency (Gleicher et al., 2013) unless the viewer has color-vision impairments or the viewer's task is exceptionally difficult (Nothelfer et al., 2017). Given anecdotal claims from some expert designers that redundant encoding can cause confusion in viewers, who typically expect color and shape to signal different nominal variables (Tufte, 1983), the lack of evidence for a perceptual advantage suggests that redundant encoding should be avoided in most cases, except when used to make visualizations accessible for viewers with color-vision impairments.

How to Design a Perceptually Efficient Visualization

Use visualizations to allow viewers to powerfully compute statistics

One core advantage of visualizations is that they capitalize on our visual system's ability to extract information